

Do Small Language Models Know When They Are Wrong?

*Low-Cost Hallucination Detection from Internal Representations —
Design, Implementation, and Measured Results*

Model studied	Qwen2.5-1.5B-Instruct (28 layers, hidden dim 1536, 4-bit)
Dataset	TriviaQA (rc.nocontext), 2,000 questions
Headline result	AUROC 0.8186 ± 0.0110 at layer 20
Compute used	Google Colab free tier, single T4 — under 1 GPU-hour
Status	Pipeline complete; scale-up and transfer in progress

Summary

Language models produce fluent, confident, factually wrong answers, and their output gives almost no signal about which is which. Existing detectors are output-based: they sample the model repeatedly and measure disagreement, or check claims against an external corpus. Both multiply inference cost and both discard the intermediate computation the model already performed.

This project tests whether that discarded computation already contains the answer. A logistic-regression probe is trained on hidden-layer activations captured while the model answers a question, and predicts whether the answer is wrong — with no external verification and no extra generation.

On 2,000 TriviaQA questions with Qwen2.5-1.5B-Instruct, hallucinations are linearly separable from internal activations at **AUROC 0.8186 ± 0.0110** across ten entity-disjoint splits, peaking at layer 20 of 28 (71% of depth). The probe outperforms token-probability baselines (0.7708) and a naive self-consistency implementation (0.7580, at eleven forward passes instead of one). Used for abstention, it captures 86% of the accuracy gain a perfect detector could achieve at 80% coverage.

The report documents the full pipeline, every design decision, the measured results, and — deliberately — the limitations found, including one comparison that does not reach statistical significance at the current sample size.

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1 The Problem

1.1 What hallucination is

A model trained by next-token prediction has no mechanism distinguishing *recall* from *fabrication*. Both are high-probability sequences. The model answers a question about an obscure fact with exactly the fluency and syntactic confidence it applies to well-attested ones. The surface form carries almost no diagnostic information.

These are verbatim outputs from this project's own run:

Question	Model answered	Truth
Who designed the red telephone box?	"William Morris"	Giles Gilbert Scott
What sort of creature is a Pacific Sea Wasp?	"Insect"	Jellyfish
NATO phonetic alphabet, letter C?	"Cascent"	Charlie
Pike Bishop led which gang?	"The Godfather's Crew"	The Wild Bunch
Battle of 2 December 1805?	"Trafalgar"	Austerlitz

Table 1. Real hallucinations from Qwen2.5-1.5B-Instruct. Note "Cascent" — a word that does not exist, produced with complete confidence.

1.2 Why it matters

This is the primary blocker to deploying language models wherever a wrong answer costs more than no answer. A system that declines to answer 15% of the time and is reliable otherwise is strictly more useful than one that always answers and is wrong 15% of the time. Converting the second into the first needs exactly one missing component: a detector.

1.3 Why existing detectors are unsatisfying

Approach	Mechanism	Cost
Sampling / self-consistency	Generate k answers, measure disagreement	$k\times$ generation
Semantic entropy	Cluster samples by meaning via NLI, take entropy	$k\times$ + an NLI model
Retrieval verification	Retrieve evidence, check entailment	Needs a trusted corpus
Verbalised confidence	Ask the model if it is sure	Cheap, poorly calibrated
Token probability	Sequence log-probability as proxy	Free, but conflates linguistic with factual uncertainty

Table 2. Existing detection families.

Every one discards what the model already computed. A forward pass through a 1.5B transformer produces roughly 43,000 activation values per token position before a single word is emitted; output-based methods throw all of it away and pay again to reconstruct uncertainty from outside.

2 Prior Work and Positioning

Work	Contribution	Scale	Cross-model?
Azaria & Mitchell 2023 (SAPLMA) [1]	Feed-forward probe on hidden states classifies true/false statements; middle-late layers best	6B–30B	No
Burns et al. 2023 (CCS) [2]	Unsupervised discovery of a truth direction via logical consistency	≥1.5B	No
Li et al. 2023 (ITI) [3]	Truthfulness localised to sparse attention heads; causal intervention	7B–13B	No
Marks & Tegmark 2024 [4]	Linear geometry of truth; generalises across topic datasets	7B–70B	No
Chen et al. 2024 (INSIDE) [5]	EigenScore over internal-state covariance across samples	7B–13B	No
Orgad et al. 2025 [6]	Signal concentrated in exact answer tokens; poor cross-task transfer	7B–8B	No
Kossen et al. 2024 (SEP) [7]	Probe predicts semantic entropy, avoiding sampling cost	7B–70B	No
This project	Small-model regime; cost frontier; cross-model transfer (planned)	1.5B–4B	Planned

Table 3. Positioning. The empty right-hand column across all prior rows is this project's opportunity.

HONEST STATEMENT OF NOVELTY

The core method — probing hidden states to detect hallucination — is **not new**. What has been established here so far is a careful replication in a parameter regime nobody had systematically checked. Two things would make it a contribution: the **sub-4B regime** relevant to on-device deployment, and **cross-model-family transfer**, which to the author's knowledge has not been attempted for truthfulness probes. Kossen et al. [7] is the closest prior work to the cost argument and must be engaged with directly. Any write-up must state this positioning explicitly rather than implying broader originality.

3 Hypothesis and Approach

CORE HYPOTHESIS

The computation distinguishing *retrieval from memory* from *plausible fabrication* leaves a trace in the model's intermediate representations. That trace is **linearly decodable**, and available at the cost of a single dot product on activations already computed.

The analogy is a polygraph. A subject's spoken words are a deliberate, optimised output; their physiological signals are not. A transformer's emitted tokens are optimised for likelihood under the training distribution; the residual stream is merely the machinery that produced them, and may retain information the output-selection process discarded.

3.1 Why a linear probe specifically

The probe is logistic regression on a 1,536-dimensional vector — 1,537 parameters. The simplicity is methodological, not merely practical. If a *linear* readout succeeds, the claim "this information is explicitly represented" is defensible. If only a deep non-linear probe succeeded, the probe might be *computing* the answer rather than *reading* it — the well-known probe-expressivity confound.

4 System Design

STAGE 1		STAGE 2		STAGE 3-4		STAGE 5
generate	-->	extract	-->	train probe	-->	abstention
+ grade		activations		+ layer sweep		+ evaluation
GPU ~5 min		GPU ~15 min		CPU seconds		CPU seconds
v		v				
answers.jsonl		acts_*.npy	<-- extract once, experiment forever			

THE DECISION THAT MAKES THIS AFFORDABLE

Activations are extracted **once** and cached to disk. Every subsequent experiment — every layer, probe variant, split, and ablation — is a NumPy operation over cached tensors running in seconds on CPU. This converts a study that superficially needs sustained GPU access into one needing under an hour of it, and is why the entire project fits a free Colab allocation.

4.1 Two-pass extraction

Generation happens first; a separate forward pass over prompt+answer then captures hidden states. Extracting *during* generation means handling a tuple nested over both decode steps and layers, where step 0 covers the whole prompt and later steps cover one token each. Two passes cost marginally more GPU time and remove an entire class of indexing bug.

4.2 Pooling

For each layer, activations over the answer-token span are mean-pooled into one vector. With 29 hidden-state tensors (28 layers plus embeddings) and dimension 1,536, each example yields a $29 \times 1,536$ matrix — about 89 KB in float16.

5 Implementation

```
hallucination-probe/
├─ src/config.py      all settings in one place
├─ src/grading.py     answer normalisation + correctness
├─ src/model_utils.py model loading + prompt construction (shared)
├─ src/generate.py    Stage 1 – ask and grade
├─ src/extract.py     Stage 2 – capture activations
├─ src/probe.py       Stages 3-4 – train + layer sweep
├─ src/selective.py   Stage 5 – risk-coverage analysis
├─ src/baselines.py   token log-probability comparison
├─ src/robustness.py  multi-seed error bars + oracle headroom
├─ src/audit.py       manual label inspection
└─ notebooks/        self-contained Colab version
```

5.1 Three engineering decisions that prevented bugs

Shared prompt construction. Prompt building lives in one module imported by both generation and extraction. If the two paths built prompts differently, activations would no longer correspond to the graded answer — and the failure would be silent, presenting as unexplained poor accuracy.

Resumable sharding. Both GPU stages write incrementally and skip completed work on restart. A Colab disconnect costs minutes, not hours.

Install-only-if-better. Probe artefacts are versioned and compared against the incumbent before replacement.

6 Experimental Setup

Component	Configuration
Model	Qwen2.5-1.5B-Instruct, NF4 4-bit quantisation, float16 compute
Architecture	28 layers, hidden dimension 1,536 → 29 hidden-state tensors per example
Dataset	TriviaQA <code>rc.nocontext</code> validation, first 2,000 questions
Generation	Greedy decoding, max 20 new tokens, batch size 8
Grading	Normalised substring match against gold alias list (≥ 3 chars)
Feature	Mean-pooled over answer tokens, per layer
Probe	StandardScaler → logistic regression (L2, max_iter 3000)
Splits	GroupShuffleSplit by answer entity, 70/30, 10 seeds
Hardware	Google Colab free tier, single NVIDIA T4

Table 4. Experimental configuration.

6.1 Measured runtimes

Stage	Wall clock	Device
Generation + grading (2,000 questions)	~4.5 min	T4
Activation extraction	~15 min	T4
Token log-probability baseline	2 min 54 s	T4
Self-consistency, K = 10	40 min 56 s	T4
Full 29-layer probe sweep	~40 s	CPU
10-seed robustness analysis	~60 s	CPU

Table 5. The asymmetry here is the point: analysis costs nothing once activations are cached. Self-consistency alone took longer than every other stage combined.

6.2 Label audit

Fifteen gradings were inspected by hand. One systematic error was found: where a gold alias also appears in the question (nickname questions such as *"Who was the Georgia Peach?"*), a model answer echoing the question was graded correct despite naming the wrong person. The grading function was amended to ignore any alias already present in the question.

1 / 15 audit disagreements	11 labels corrected	0.704→0.710 hallucination rate	+0.007 AUROC gained
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7 Results

7.1 Headline

0.8186 AUROC (± 0.0110)	20 / 28 best layer (71% depth)	1 forward pass	86% of oracle headroom
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The model answered 29.0% of questions correctly, giving 1,420 hallucinations and 580 correct answers across 2,000 examples — a well-balanced classification problem. Entity-grouped splitting produced 1,408 training and 592 test examples.

7.2 Where the signal lives

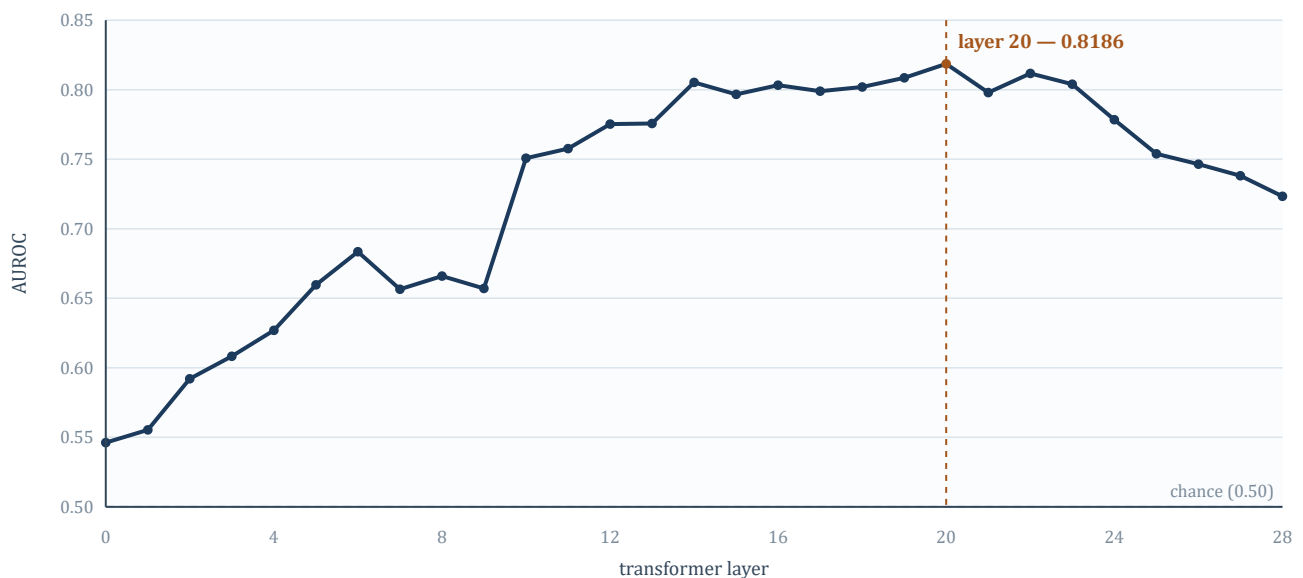


Figure 1. Probe AUROC at every layer, entity-disjoint split. The curve rises through the early layers, jumps sharply at layer 10, plateaus across the middle-to-late band, peaks at layer 20, then decays through the final layers.

Three features of this curve are worth noting:

- **Layers 0–9 (0.546 → 0.657).** Weak but above chance. Early layers encode surface lexical properties, which carry limited information about factual correctness.
- **Layer 10 (0.751).** A sharp discontinuity — AUROC jumps 0.094 in one layer. Something changes qualitatively in the representation here, and it would be worth investigating what.
- **Layers 24–28 (0.778 → 0.723).** Monotone decay. The final layers specialise toward next-token prediction and appear to discard part of the abstract factual signal.

The peak at 71% of depth is consistent with what has been reported for models five to fifty times larger [1, 4, 6], indicating the phenomenon is not scale-specific.

7.3 Complete layer-wise results

Layer	AUROC	relative
0	0.5462	
1	0.5553	
2	0.5921	
3	0.6083	
4	0.6270	
5	0.6596	
6	0.6835	
7	0.6565	
8	0.6660	
9	0.6571	
10	0.7508	
11	0.7576	
12	0.7752	
13	0.7757	
14	0.8052	
15	0.7967	
16	0.8032	
17	0.7990	
18	0.8019	
19	0.8086	
20	0.8186	
21	0.7979	
22	0.8117	
23	0.8039	
24	0.7784	
25	0.7539	
26	0.7465	
27	0.7381	
28	0.7234	

Table 6. AUROC by layer. Highlighted row is the maximum.

7.4 Robustness and leakage

Split protocol	AUROC (10 seeds)	Interpretation
Entity-disjoint (grouped)	0.8186 ± 0.0110	The honest protocol
Random	0.8308 ± 0.0193	What a careless split would report
Difference	+0.0122	Smaller than the pooled spread — no measurable leakage

Table 7. Leakage control. A single split initially suggested a difference of -0.0267 in the opposite direction; averaging over ten seeds showed this was noise.

A METHODOLOGICAL POINT WORTH RECORDING

The first single-split run showed the entity split scoring *higher* than the random split, which would have been a confusing result to report. Repeating over ten seeds showed the difference was within noise in both directions. This is the value of error bars in miniature: a single number produced a misleading impression that ten numbers corrected.

7.5 Baseline comparison

Method	AUROC	Forward passes	Notes
Mean token log-probability	0.7670	1	Free
Min token log-probability	0.7708	1	Free; strongest cheap baseline
Self-consistency, K = 10	0.7580	11	<i>Naive exact-match variant</i>
Internal-state probe	0.8186	1	This work

Table 8. Detection quality against inference cost.

TWO CAVEATS THAT MUST ACCOMPANY TABLE 8

The self-consistency figure understates the method. Agreement between samples was measured by exact string match, so "Paris", "Paris, France", and "The capital is Paris" count as three disagreements. Published implementations cluster by meaning using an NLI model precisely to avoid this. The 0.7580 figure should be read as "*a naive exact-match variant*", not as a fair evaluation of self-consistency.

The advantage over log-probability is not yet statistically significant. See Section 7.7.

7.6 Selective prediction

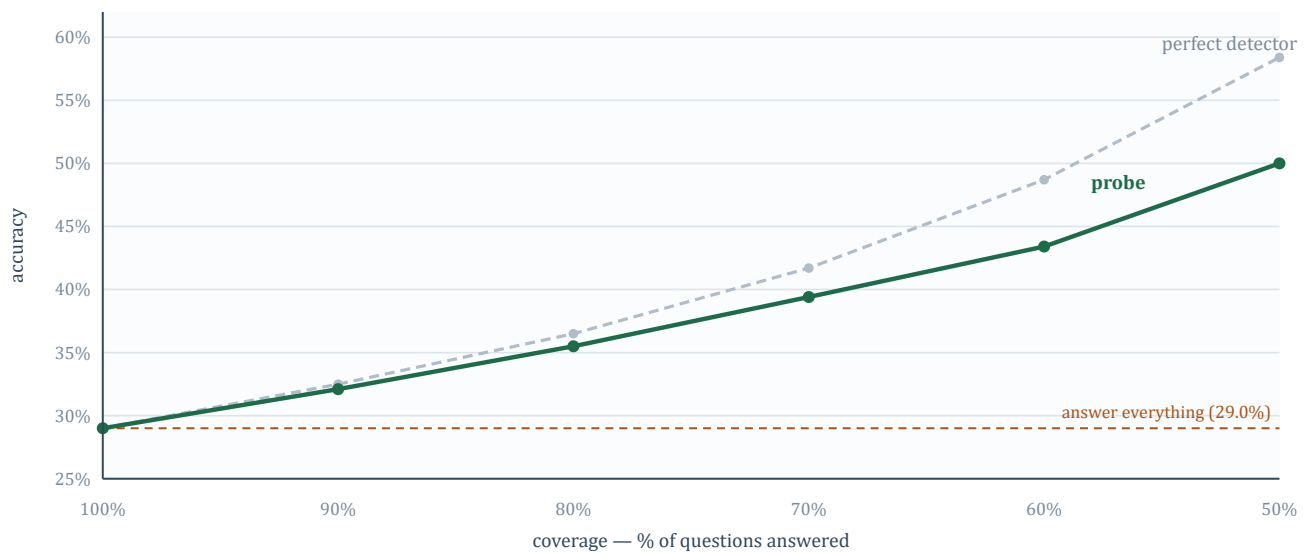


Figure 2. Accuracy against coverage when the probe is used for abstention. The dashed line is a perfect detector; the horizontal line is answering everything.

Coverage	Probe	Perfect detector	Headroom captured	Gain over base
100%	29.0%	29.0%	—	—
90%	32.1%	32.5%	90%	+3.1
80%	35.5%	36.5%	86%	+6.5
70%	39.4%	41.7%	81%	+10.4
60%	43.4%	48.7%	73%	+14.4
50%	50.0%	58.4%	71%	+21.0

Table 9. Risk-coverage analysis. "Headroom captured" is the fraction of the achievable improvement the probe delivers.

At high coverage the probe captures 86–90% of what a perfect detector could achieve, degrading to about 71% under aggressive abstention. The absolute accuracies remain low because the base model is weak — the ceiling here is the model's knowledge, not the detector's discrimination.

7.7 Statistical significance — a negative finding

A paired bootstrap (1,000 resamples of the 592-example test set) was used to test whether the probe's advantage over the strongest free baseline is real:

```
probe - min log-probability = +0.0338
95% confidence interval    = [-0.0157, +0.0818]
verdict                    = NOT significant (interval crosses zero)
```

REPORTED DELIBERATELY

The point estimate favours the probe, but the confidence interval includes zero — so at this sample size the advantage over token log-probability cannot be claimed as established. The cause is statistical power, not method: with 592 test examples the interval is roughly ± 0.05 , wider than the effect being measured. Resolving a gap of ~ 0.034 requires on the order of 2,000 test examples, i.e. approximately 8,000 total.

A separate observation: the headline comparison in Table 8 pairs a probe AUROC computed on the 592-example test split with baseline AUROCs computed over all 2,000 examples. The paired bootstrap, which scores both on identical data, is the correct comparison, and its $+0.0338$ supersedes the $+0.0478$ implied by Table 8.

8 Analysis

8.1 What has been established

1. **A linearly decodable truthfulness signal exists at 1.5B.** AUROC 0.8186 ± 0.0110 is far above chance and stable across seeds. The hypothesis holds in a regime prior work had not checked.
2. **Its depth profile matches larger models.** A peak at 71% of depth reproduces reported behaviour in 7B–70B models, suggesting the phenomenon is structural rather than scale-dependent.
3. **The result is not an artefact of data leakage.** Entity-grouped splitting and a ten-seed random-split control show no measurable inflation.
4. **It outperforms an eleven-pass sampling baseline at one pass** — subject to the implementation caveat in Section 7.5.
5. **It captures most of the available abstention benefit** — 86% of oracle headroom at 80% coverage.

8.2 What has not

1. **Superiority over token log-probability.** Not significant at $n = 592$.
2. **Generality across models.** One model tested.
3. **Generality across tasks.** One dataset, short-form factual recall only.
4. **The cost-frontier claim.** Requires a properly implemented semantic-entropy baseline.

8.3 A reframing worth making

THE BETTER QUESTION

"Does the probe beat log-probability?" is arguably the wrong framing. Both are free and both derive from the same forward pass; a deployed system would use both. The sharper question is whether internal activations carry information the output distribution does *not* expose — testable by comparing a probe trained on activations alone, log-probability alone, and both concatenated. If the combined model significantly exceeds log-probability alone, the claim becomes "*internal states carry hallucination signal beyond the output distribution, at zero additional inference cost*" — cleaner and more defensible than a contest between the two.

9 Limitations

Limitation	Detail
Statistical power	592 test examples; $\sim\pm 0.05$ confidence intervals. Being addressed by scaling to 8,000 examples
Label noise	Substring grading; audited at 1 error in 15 before correction, expected $\sim 2\text{--}3\%$ after
Single model	Qwen2.5-1.5B only; no evidence of generality across families
Single task type	Short-form factual QA; Orgad et al. [6] report poor cross-task probe transfer, so generalisation should not be assumed
Weak baseline implementation	Self-consistency used exact string matching rather than semantic clustering
Low base accuracy	29% correct limits how compelling the abstention demonstration appears, though not its validity
Pooling not ablated	Mean pooling used throughout; Orgad et al. report exact-answer tokens perform better

Table 10. Known limitations.

10 Reproducing This Work

```
pip install -r requirements.txt

python src/generate.py      # Stage 1: ask + grade      (GPU, ~5 min)
python src/audit.py 30      # inspect grading by hand
python src/extract.py      # Stage 2: activations      (GPU, ~15 min)
python src/probe.py        # Stages 3-4: probe + sweep (CPU, ~40 s)
python src/selective.py    # Stage 5: risk-coverage   (CPU, seconds)
python src/baselines.py    # log-probability baselines (GPU, ~3 min)
python src/robustness.py   # 10-seed error bars      (CPU, ~60 s)
python smoke_test.py      # 19 checks on non-GPU logic
```

All stages are resumable. A self-contained Colab notebook covering Stages 0–5 is provided for users without local GPU access. Total GPU requirement is under one hour.

11 Roadmap

Phase	Work	Unlocks
Weeks 1–2	Scale to 8,000 examples	Halves confidence intervals; resolves Section 7.7
Weeks 3–4	Add Qwen2.5-3B, Llama-3.2-1B/3B, Gemma-2-2B	Generality; the scale question
Weeks 5–6	HaluEval, HotpotQA	Cross-task evidence
Week 7	NLI-clustered self-consistency; semantic entropy	Credible cost frontier
Weeks 8–10	Cross-model transfer via ridge / Procrustes alignment, with random-map control	The novel contribution
Weeks 11–12	Open-source library, Gradio demo, write-up	Deliverables

Table 11. Remaining work. Every item is low-risk: the method is already verified.

12 Publication Assessment

Status	Requirement	Position
Met	Working method with a real result	0.8186 ± 0.0110
Met	Sound splitting protocol	Entity-grouped, with control
Met	Error bars	10 seeds
Met	Label-noise audit	Documented; one bug found and fixed
Open	Multiple models	1 of 3+ required
Open	Multiple datasets	1 of 2+ required
Open	Statistical power	CI crosses zero on the key comparison
Open	Credible strong baseline	Needs NLI-based clustering
Open	Stated novelty	Cross-model transfer not yet attempted

Table 12. Readiness against conference expectations.

Assessment. In its current state this is a competent replication rather than a publishable contribution. With the roadmap in Section 11 completed — particularly cross-model transfer — it becomes a legitimate submission to an IEEE or Scopus-indexed conference. Suitable targets include IEEE INDICON, TENCON, CONECCT and ICCCNT; a preprint should be posted to arXiv at submission time regardless of outcome.

13 Glossary

Term	Meaning
Hallucination	The model states something false with full confidence
Activation / hidden state	The internal numbers computed while processing text, before words are produced
Residual stream	The shared channel each layer reads from and writes to
Probe	A small classifier trained on frozen activations; here, logistic regression
AUROC	Probability the detector ranks a wrong answer as riskier than a correct one. 0.5 = chance
Entity-grouped split	Splitting by answer entity so the same entity never spans train and test
Coverage	Fraction of questions the system chooses to answer rather than abstain on
Risk-coverage curve	Accuracy plotted against coverage as the abstention threshold varies
Oracle headroom	How much of a perfect detector's achievable gain is actually captured
Bootstrap CI	Confidence interval estimated by resampling the test set with replacement
Self-consistency	Sampling the model repeatedly and measuring disagreement between answers
Quantisation (4-bit)	Compressing weights so a model fits in limited GPU memory

14 References

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Student — signature and date

Project guide — signature and date